

# An efficient low power method for FinFET domino OR logic circuit

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## ABSTRACT

Advancement in the metal oxide semiconductor field effect transistor (MOSFET) technology leads to large power dissipation, and propagation delay due to extreme scaling of the transistor. So, very large-scale integration (VLSI) industry requires alternative options to meet the demands of the customers. Fin-shaped field effect transistor (FinFET) is new innovative device to replace the MOSFET technology in ultra-deep submicron (ultra-DSM) regime. FinFET mitigates power dissipation to the large extent as compared to the MOSFET, but still the impact of power dissipation degrades the circuit performance. Leakage reduction techniques are further needed in ultra-DSM regime to reduce the power dissipation. Domino logic is commonly used in large memories, and high-speed processors. In this article, a low power domino logic is proposed using FinFET devices. The proposed method is compared to the existing similar domino logic approaches for the different performance metrics at 7 nm technology node using Cadence's tools. The proposed method effectively reduces the leakage power dissipation, and delay penalty for the different logic circuits. The reliability of the domino OR logic is checked by estimating the temperature, and multiple parallel fins variations. The proposed method has a high noise tolerance capacity in term of unity noise gain (UNG) for the domino OR logic as compared to the existing approaches.

## 1. Introduction

Complementary metal oxide semiconductor (CMOS) is one of the revolutionary inventions in the field of VLSI. Scaling of MOSFET results in miniature of electronic systems, which provides high performance operations. Electronic industry has achieved many milestones in CMOS technology, which incorporates different engineering fields to provide the advancement in the different domains [1]. High integration of the devices is possible if the size of the transistor is small. Small size transistor elevates large leakage current due to short channel effect (SCE). Key obstacle in the advancement of the CMOS technology is SCE, which degrades the performance of the circuits. In ultra-DSM regime, leakage current starts to overlook the performance of the CMOS circuits. The main concern in ultra-DSM regime is to mitigate the leakage power dissipation, because it affects the various parameters of the transistor. Leakage power dissipation is observed as shown in Eq. (1).

$$P_{leakage} = V_{DD} \times I_{leakage} \quad (1)$$

Here,  $V_{DD}$  is power supply voltage, and  $I_{leakage}$  is the total leakage current.

$V_{DD}$  is fixed for a particular technology node. So, leakage current plays a vital role to control the leakage power dissipation. Leakage

power dissipation can be mitigated by varying the various transistor's parameters. The various transistor's parameters are affected the leakage current as expressed in Eq. (2).

$$I_{leakage} = \mu_0 C_{OX} \frac{W}{L} e^{1.8} V_T^2 e^{\frac{(V_{GS} - V_{TH} - \eta V_{DS})}{mV_T}} \left(1 - e^{-\frac{V_{DS}}{V_T}}\right) \quad (2)$$

Where,  $V_{TH}$  represents threshold voltage,  $W/L$  is the ratio of channel width to the effective channel length,  $C_{OX}$  denotes gate oxide capacitance per unit width,  $\mu_0$  indicates zero bias mobility,  $V_T$  represents thermal equivalent voltage,  $m$  depicts sub-threshold swing coefficient,  $V_{DS}$  is drain to source voltage, and  $\eta$  is drain induced barrier lowering coefficient.

Leakage current depends on various transistor's parameters as given in Eq. (2). Therefore, leakage reduction technique utilizes the different controlling methods by using these parameters [2]. The different transistor arrangements can alter the performance of the circuits. It means that the robustness of a cell can be increased by selecting the most suitable topology. To enhance the performance of the circuits, researchers introduced various mechanisms in which bulk of the transistors are connected differently. In logic circuits, high driving capability can be achieved by lowering the threshold voltage of ON-transistors, and less leakage current is maintained by the high threshold voltage of

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OFF-transistors. The main goal is to reduce the wastage of power in logic circuits. In transistors, different arrangements of connections between the bulk and gate terminals can provide the required performance. Threshold voltage of transistor is altered as shown in Eq. (3) [3].

$$V_{TH} = V_{TH0} - \gamma \left( \sqrt{|-2\phi_F|} - \sqrt{|-2\phi_F| + V_{BS}} \right) - \eta V_{SD} \quad (3)$$

Where,  $\gamma$  is body bias coefficient,  $V_{TH0}$  is threshold voltage at  $V_{BS} = 0$ ,  $2\phi_F$  is silicon surface potential, and  $V_{BS}$  is body to source voltage.

If the bulk terminal of the transistors is connected differently, then it may improve the performance of the circuits. In dynamic threshold technique, bodies of the transistors are connected to at different voltage levels. It may produce low leakage current. Different transistors arrangements are also providing fast operations, strong robustness against transient, and permanent faults. It also mitigates the bias temperature instability effects. These types of techniques minimize the transistor count in circuits that decreases power consumption, and delay. In VLSI circuits, same logic function can be obtained from different type of transistors networks [4]. These types of distinct transistors arrangements provide different physical, and electrical characteristics. There are the different possibilities for choosing the best topology for implementing the logic cells with the different transistors network in such a way that it provides less leakage, minimum area overhead, and mitigates process variations.

Aggressive scaling of the MOSFET leads to SCE, which increases leakage current and starts to affect the circuit performance. Therefore, researchers innovate a new device known as FinFET to minimize the leakage power dissipation in ultra-DSM regime [5]. FinFET is a three-dimensional (3D) multi-gate (MG) transistor, which has better controllability, isolation, less fluctuations, higher switching speed, less wafer area, improved electrostatic control, better carrier mobility, and better trans-conductance as compared to MOSFET device. The best part of the FinFET is 3D geometric structure, which contains drain, source, front gate (FG) and back gate (BG) terminals. This exclusive structure of the FinFET is the reason behind the great performance.

Double gate structure of the FinFET provides better control on the channel length, which helps to reduce the SCE and minimizes the leakage current. Both gates of the FinFET can be utilized independently to supervise the threshold voltage, and current. This type of configuration is known as independent gate (IG) FinFET. The other type of the configuration is known as shorted gate (SG) FinFET, where both the FG and BG are shorted together and fully covered by the fin from the three sides. Fig. 1 illustrates IG and SG FinFET configurations.

In SG FinFET circuit, both gates are biased together which provides maximum drive current, and less area overhead. Main drawback of this configuration is less control on the threshold voltage of the transistor. On the other hand, IG FinFET provides better control on threshold voltage, and has more area overhead due to more space between the source and drain terminals. In IG FinFET configuration, both gates are biased separately and offers less off-state current as compared to SG FinFET. Most prominent feature of FinFET is its unique 3D geometrical

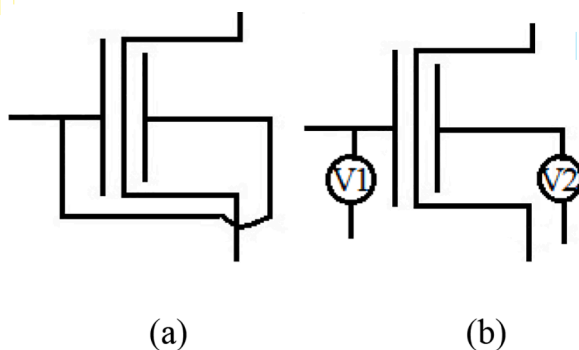


Fig. 1. Symbols of (a) SG FinFET (b) IG FinFET.

structure, which helps to improve the circuit performance [6,7]. Fig. 2 depicts the basic FinFET structure.

Fig. 2 illustrates the geometric and 3D structures of FinFET. FinFET is a non-planar, three-dimensional, dual-gate transistor fabricated on a silicon substrate that allows better leakage current regulation. Since fin height is greater than fin width therefore, many parallel fin structures can be used in FinFET for a large fin width. Fin width is the vital parameter in finFET structure, which helps in scaling of the transistor. Scaling factor is symbolize by  $\lambda$  and given in Eq. (4) [8].

$$\lambda = W_{fin} + 2t_{ox} \quad (4)$$

From Eq. (4), it can observe that scaling factor depends upon fin width ( $W_{fin}$ ) and oxide thickness ( $t_{ox}$ ). The fin width of the FinFET is calculated as shown in Eq. (5) [9].

$$W_{fin} = 2H_{fin} + t_{fin} \quad (5)$$

Where,  $W_{fin}$  represents width of the fin,  $t_{fin}$  denotes fin thickness and  $H_{fin}$  is the height of the fin. The effective width of the device contains  $n$  number of fins and shown in Eq. (6) [10].

$$W_{eff} = nW_{fin} \quad (6)$$

Where,  $n$  represents identical fins in parallel. The Eq. (7) is used to calculate the effective channel length of a FinFET device [11].

$$L_{eff} = 2L_{fin} + L_g \quad (7)$$

Where,  $L_{fin}$  is the length of the fin on both sides, and  $L_g$  is the length of the gate that covers the fin.

Nowadays, the different logic circuits are implemented using FinFET devices for low-power, and high-speed applications. Inverting, and non-inverting types of hybrid line decoders were developed using FinFET technology for low-power, and high-speed performance [12]. The developed decoder contains a transmission gate, modified gate diffusion input, and NAND logic. The circuit simulations for the power, speed, and PDP parameters were performed for CMOS and FinFET-based technologies, and compared the results at different frequencies, and power supplies. A low-power, and high-speed 7-transistor static random-access memory was designed using FinFET technology [13]. Leakage power was minimized using self-controllable voltage level approach. The leakage power was further reduced using lower, upper, and mixed form of lower-upper self-controllable voltage level approaches. The primary idea was to put the access transistors in OFF-state during the standby mode of the cell. Low-transistor count based approximate compressors of different accuracy, and performance characteristics were investigated using FinFET devices [14]. The different approximate multipliers were developed for the different hybrid arrangements using the different investigated approximate compressors. The effectiveness of the investigated approximate multipliers was also checked for the neural network, and the image processing applications.

Domino logic gates are the most powerful logic gates for the massive memories, and high-speed processors as compared to CMOS logic gates. Domino logic gates have low noise margin, which allows them to operate at full speed, but it also makes them more sensitive to noise effects. The most powerful way to improve the circuit's noise immunity is to scale down the technology. But, it rises the circuit's power dissipation. Supply voltage and threshold voltage of the device must be scaled to ensure the successful trade-off between the noise immunity, and power dissipation. Threshold voltage, leakage current, and noise sources are the primary causes for the domino circuit's output performance. Footer domino logic (FDL), and footer-less domino logic (FLDL) are the two types of the domino configurations. FDL configuration added one extra transistor in footer side, which helps to control the unwanted charge in the path from dynamic node to ground. FLDL configuration saves the implementation area. The schematic diagram for the FLDL configuration is presented in Fig. 3 [15].

In domino logic, the clock signal is the crucial component. The

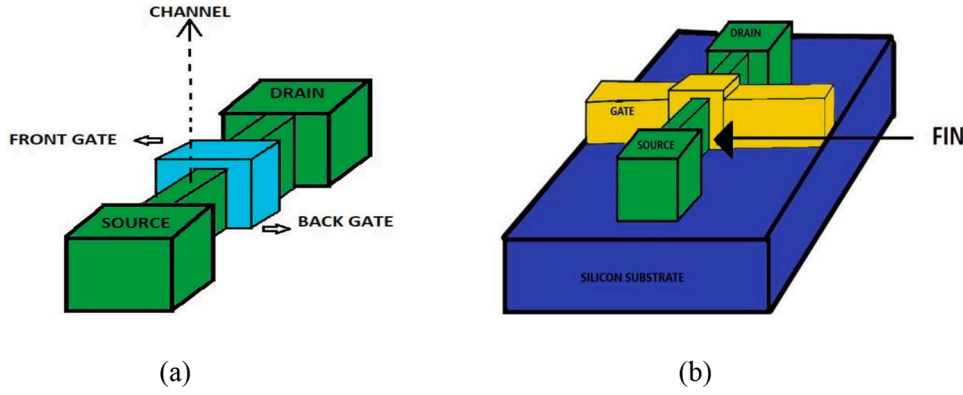


Fig. 2. FinFET structure representation (a) geometric (b) 3D.

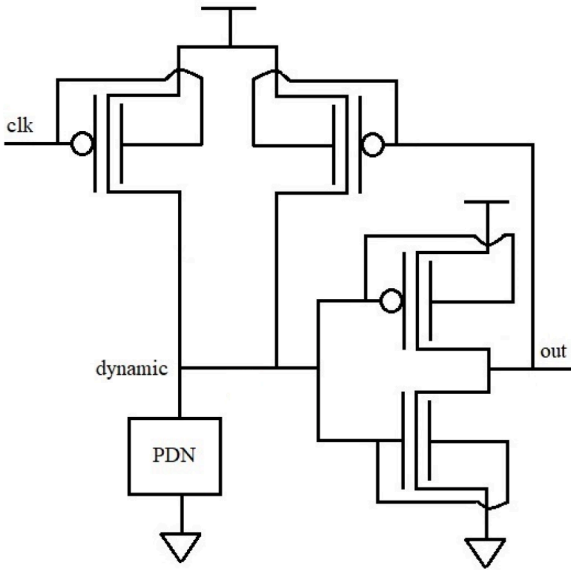


Fig. 3. Schematic diagram for FLDL logic.

operation of the domino logic circuits depends on the nature of clock signal. Pre-charge and evaluation are two phases in domino logic. Pre-charge phase operates if clock signal is at logic low, while evaluation phase activates if clock signal is at logic high. Since the dynamic node is continuously charged to VDD via the pre-charge transistor, when the clock signal is low. So, the output node does not shift and remains at logic low. The output node of the domino logic is varied according to the primary inputs added to the pull-down network (PDN) during the evaluation phase only. The output of the domino circuits can be aided by using a keeper transistor. Keeper transistor overcomes the problem of charge sharing between the transistor's junction capacitances. To avoid the charge sharing, a keeper transistor is mounted in parallel with the pre-charge transistor, and its gate is attached to the output node [16].

Since, FinFET has a range of advantages over the widely used CMOS technology. In this research paper, we have introduced a new FinFET-based low power domino logic for the improvement of the output's performance. The suggested method is validated for the different domino logic circuits such as buffer, NOT, AND and OR gates. Leakage power dissipation, propagation delay, power delay product (PDP), temperature variations, fins variations and unity noise gain (UNG) metrics are considered for the performance estimation. The simulation results are obtained at 7 nm technology node using Cadence's tools. The proposed method is compared with the other available domino logic approaches for the different performance metrics to demonstrate the efficiency of the proposed approach.

The rest of the paper is organized as follows: Previous related works are discussed in Section 2. Section 3 provides the proposed low power domino logic method. The simulation results for different logic circuits are discussed, and compared in Section 4. Section 5 concludes the paper.

## 2. Previous related work

In ultra-DSM regime, leakage power dissipation is a serious problem that affects the overall performance of the logic circuits. Leakage power dissipation increases dramatically with the advancement in the technology [17]. Researchers have suggested a number of leakage reduction strategies to minimize the impact of leakage power dissipation while retaining the circuit's performance. FinFET device produces less power dissipation as compared to MOSFET device. But, the power dissipation cannot be neglected in the ultra-DSM regime. Many approaches have been applied by the researchers to reduce the impact of the power dissipation, and each technique offers a less power dissipation with some trade-off [18]. Many researchers had introduced the various transistor level leakage reduction techniques, which can be incorporated in different circuits to reduce the power dissipation. The latest and important domino logic approaches are discussed in this section.

Researchers suggested a stacking effect model to reduce the power dissipation. According to this model, if more than one transistor connected in series are switched-off, then logic circuit will produce less power dissipation. The stacking effect model is commonly used for reducing the power dissipation because of its easy arrangement and operation.

Leakage control transistor (LECTOR) technique was proposed to address the problem of leakage current [19]. LECTOR technique forms the transistors stack by integrating two additional leakage control transistors (LCTs) in a CMOS logic. One extra PMOS transistor is attached to the pull-up network (PUN), and one extra NMOS transistor is connected to the PDN. LECTOR technique employs the stacking effect, and does not require any external circuitry to operate the LCTs. LECTOR technique reduces the leakage current effectively, and works well in both active and idle phases of the circuits. In different circuits, CMOS logic cell consists of various series and parallel combinations of NMOS, and PMOS transistors. The total leakage current contributes by the parallel connected transistors is the sum of the currents flowing through each transistor. However, the evaluation of the leakage current through the series connected transistors is quite complex due to its non linear characteristics. To evaluate the current flowing through stack of the three transistors during subthreshold region is given by Eq. (8) [19].

$$I_{sub1} : I_{sub2} : I_{sub3} = 1.8 \exp\left(-\frac{\eta V_{DD}}{\eta V_t}\right) : 1.8 : 1 \quad (8)$$

Where,  $I_{subi}$  ( $i = 1, 2, 3, \dots$ ) is subthreshold leakage current for  $i^{th}$  stacked transistor. If there are more than three stacked transistors in a circuit, the leakage current is very minimal, and can be ignored. The

motive behind this technique is to mitigate leakage power by stacking of the transistors in the path from supply voltage to ground.

The primary requirement for any logic circuit is its input-output response. The input-output response of the LECTOR technique is poor, which skewed the output level. The small variation in the output level will influence the performance of the circuits in ultra-DSM regime because the voltage levels are small. In LECTOR technique, distorted output level occurs due to the improper charging and discharging of the intermediate nodes. LECTOR with footed diode inverter domino logic was developed by mixing the concepts of LECTOR and domino logic [20]. Authors had simulated 2-input AND, OR, 4-input OR and 8-input OR domino circuits at 45 nm technology node using BSIM4 model file. A conventional FDL domino logic was applied for the comparison purpose. The schematic arrangement for the LECTOR with footed diode inverter approach is shown in Fig. 4.

Main limitation of LECTOR technique in ultra-DSM regime is inappropriate output waveform. So, in ultra-DSM regime, leakage reduction techniques with appropriate output logic is preferred. On/OFF logic (ONOFIC) approach is an effective method to get the accurate output. ONOFIC technique improves the efficiency of the circuits by delivering the correct logic level at the output node [21].

Two additional transistors are exercised in between the PUN, and PDN of the CMOS logic to limit the leakage current. This additional module is called ONOFIC block. PMOS transistor in ONOFIC block controls the operation of NMOS transistor. Depending on the output logic, ONOFIC NMOS transistor will be ideally in OFF-state or ON-state. ONOFIC is circuit level technique, and main goal of this approach is proper ON/OFF property. This approach provides the accurate logic level at the output node. It utilizes the same  $V_{TH}$  throughout the circuit, which eliminates the different threshold voltage control mechanisms. Transistors stacking is the idea to mitigate the leakage power in ONOFIC approach. When the circuit is in the OFF-state, then ONOFIC block regulates the leakage current, and provides the maximum resistance. It

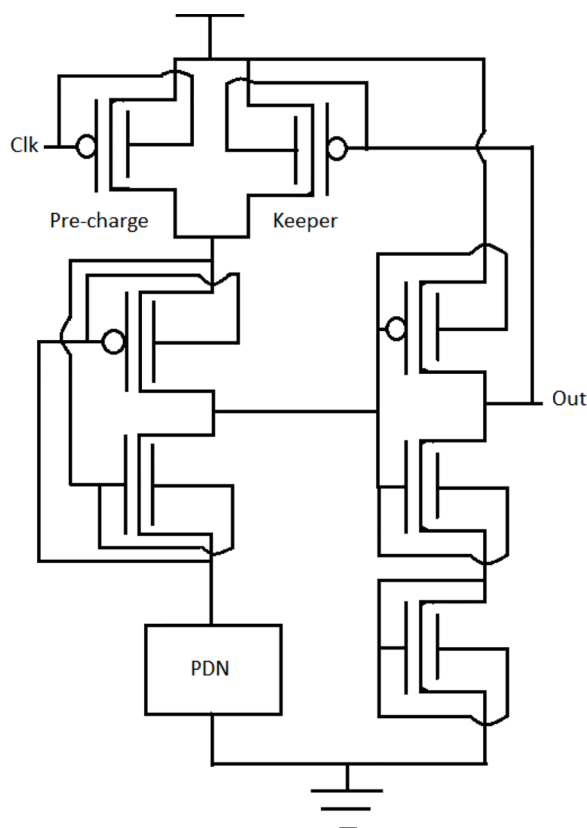


Fig. 4. Schematic diagram for LECTOR with footed diode inverter approach.

is also observed that propagation delay of the ONOFIC circuits are less than LECTOR circuits. Therefore, ONOFIC approach improves the speed of the circuits. ONOFIC block has a shorter turn-ON, and turn-OFF times due to the quicker operation of the PMOS transistor, and so has a shorter propagation delay than other approaches.

The explicit ON/OFF functionality of the ONOFIC block improves the input-output response. An ONOFIC pull-up approach was developed for the domino logic, where the ONOFIC block was inserted in the PUN [22]. Authors had simulated 2-input AND, OR, 4-input AND, OR, 8-input OR and 16-input OR domino circuits at 32 nm technology node using FinFET devices. The schematic diagram for the ONOFIC pull-up method is provided in Fig. 5.

The alternate ONOFIC is an extended version of the ONOFIC approach that uses two ONOFIC blocks between the PUN and PDN to minimize the significant leakage current [23]. Two path transistors are used in between the PUN and PDN of the alternate approach, while two forced transistors are used to monitor the path transistor's inputs. For any input combination, one ONOFIC block is switched-on, and the other is turned-off at a time. This method provides less power dissipation with the cost of propagation delay. A domino logic was constructed using the alternate ONOFIC method [24]. Authors had simulated 2-input AND, OR, 4-input OR, 8-input OR and 16-input OR domino circuits at 32 nm technology node using FinFET devices. The schematic diagram for the alternate ONOFIC domino logic is depicted in Fig. 6.

Extra transistors are required in domino logic for appropriate functionality of the circuits. The dynamic node driven feedback transistor domino logic (DNDFTDL) technique was proposed for the domino circuits to minimise the power dissipation, and noise element [16]. DNDFTDL technique was proposed for the FDL configuration. It contains three extra transistors as compared to conventional FDL configuration. Fig. 7 illustrates the schematic diagram for the DNDFTDL technique.

When the clock signal is low during the pre-charge phase, the pre-charge transistor turns ON. The dynamic node charges to a high level value. It produces a low output value due to presence of an inverter circuit. The keeper transistor is turned ON by the output node, which avoids unwanted charge leakage. The clock signal is high during the evaluation phase. The pre-charge transistor will be turned OFF during

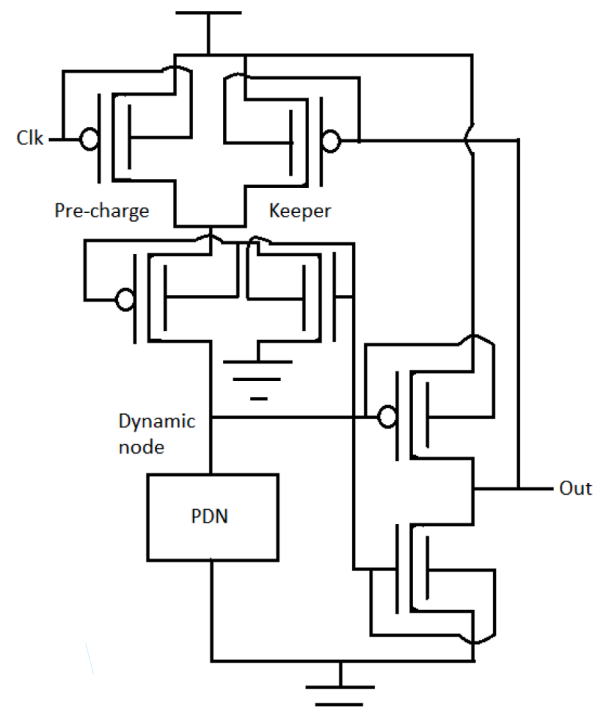


Fig. 5. Schematic diagram for ONOFIC pull-up approach.

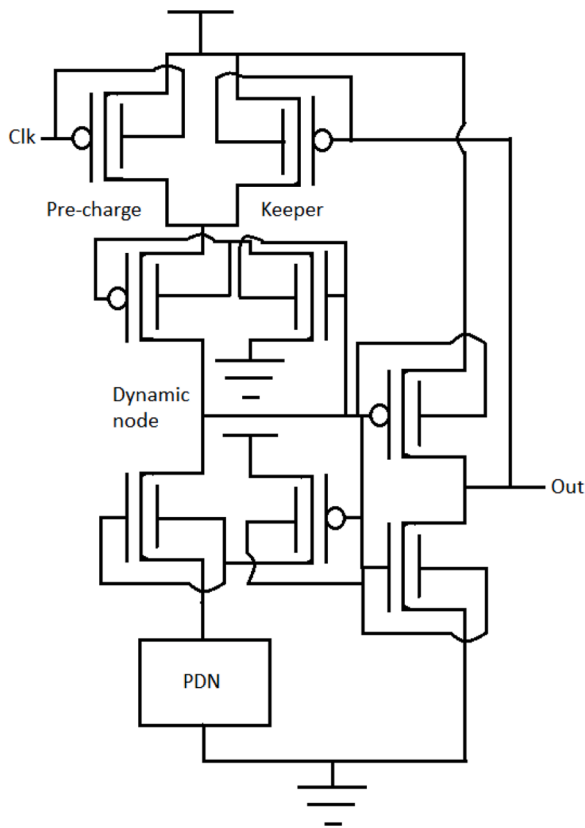


Fig. 6. Schematic diagram for the alternate ONOFIC domino approach.

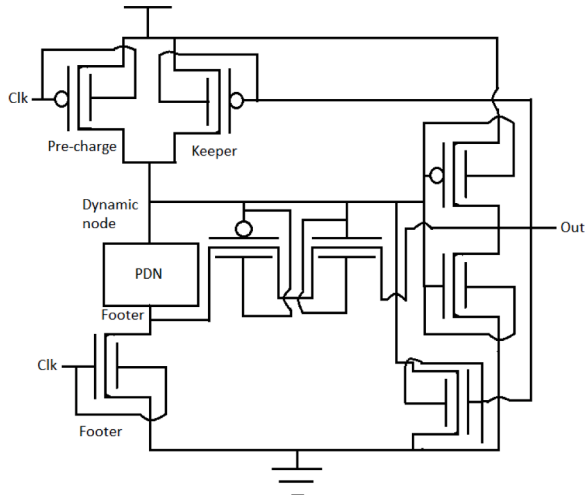


Fig. 7. Schematic arrangement for DNDFTDL technique.

evaluation phase, and the footer transistor will be turned ON.

The n-channel transistor arrangements in PDN determine the discharging of dynamic node to the ground. The presence of noise produces a voltage drop across the footer node, and lowers the gate to source voltage of evaluation logic. Two transistors are connected in series form between the footer node to the output node to prevent the leakage current. It creates a high-resistance between the two nodes. In some cases, an extra transistor is necessary in domino logic for the circuit to function properly.

Foot driven stack transistor domino logic (FDSTD) technique was established to improve the domino circuit's performance. FDSTD technique was developed by using the FDL idea [8].

FDSTD circuit comprises two parts: an input part with a pre-charge transistor, and PDN with n-channel transistors. The output section includes inverter logic, keeper transistor, and two stacked n-channel transistors. The circuit runs in the pre-charge phase, when the clock signal is low. If the pre-charge transistor is turned ON, the dynamic node is charged to  $V_{DD}$ . Due to inverter logic, the output node is at low logic value. The keeper transistor is turned ON by the output signal to avoid unwanted charge leakage through the dynamic node. Both stacked n-channel leakage control transistors are turned OFF during this phase. It prevents the dynamic node from being discharged.

FDSTD logic enters in the evaluation phase when the clock signal is high. The pre-charge transistor is turned OFF, while the footer transistor is turned ON during this phase. Dynamic node discharges through PDN depending on the arrangement of the n-channel transistors, and produces high logic at the output node. The footer node is in a low state during this state, which turns OFF the n-channel transistor attached to it. There will be no leakage between the dynamic node and the ground. The schematic diagram for FDSTD technique is shown in Fig. 8.

FDSTD schematic consists a FDL configuration with two additional transistors. The authors modelled and simulated various input domino OR gates using FinFET devices at 32 nm technology node.

In this paper, a new low power domino logic is proposed, and compared with the existing techniques as discussed in this section to show the reliable, and improved performance of the proposed logic.

### 3. Proposed method

Domino logic circuit requires less implementation area, and processing the high speed as compared to the conventional CMOS technology. Hence, it is widely employed for making the high-speed memory processors. Therefore, many researchers had worked at different abstraction levels for the development of the effective domino logic.

In this section, an efficient low power domino logic is proposed at circuit level design using minimum number of the transistors. FinFET device exhibits minimum leakage current as compared to the MOSFET device. So, FinFET devices are used for the proposed method. SG FinFET devices are preferred here because of simple, and fixed arrangement of the gate terminals. A schematic diagram for the proposed low power domino method is shown in Fig. 9.

Three additional devices are inserted in the proposed low power

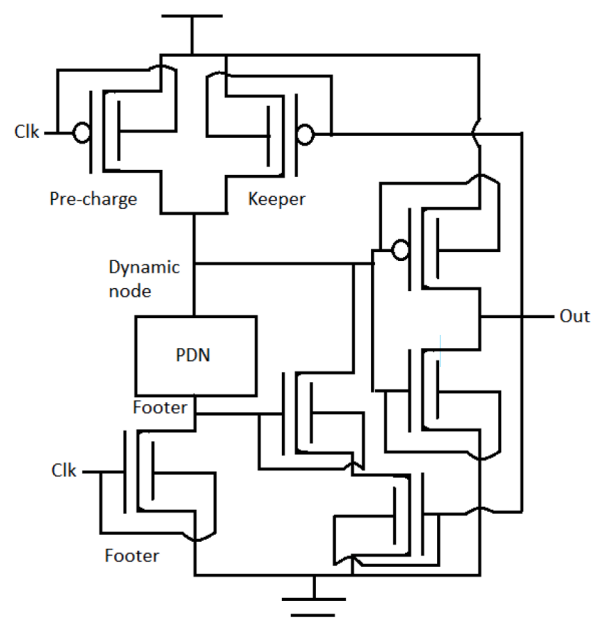


Fig. 8. Schematic diagram for FDSTD technique.

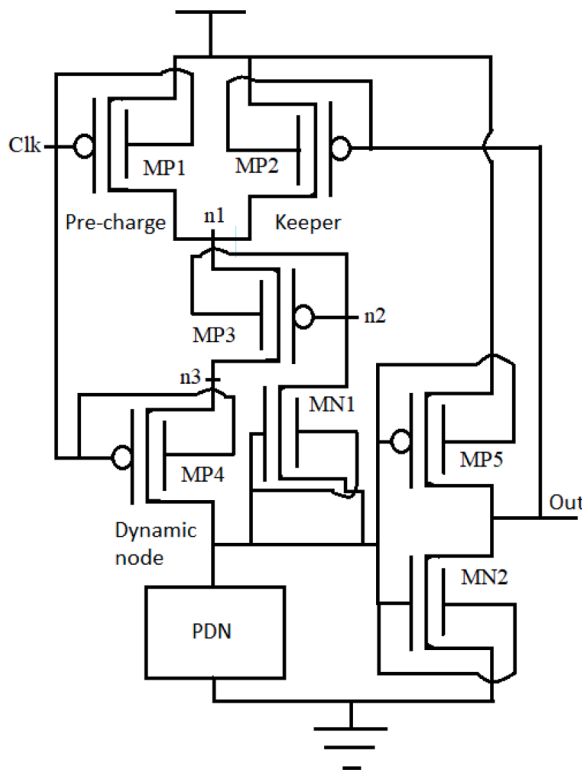


Fig. 9. Schematic diagram for the proposed low power domino method.

domino logic as compared to the conventional FLDL configuration. The proposed method is a fixed configuration of the transistors. That avoids the dependency of any parameter of the transistor on the operating status of the domino logic circuits. The proposed method is a single threshold voltage approach, which overcomes the issue of different threshold voltage implementations. MP1 is pre-charge transistor, while MP2 is a keeper transistor. MP3 and MP4 forms the transistors stack and help to reduce the large leakage path. The operation of the MP3 depends on the MN1 transistor. The operating nature of the MP3 transistor provides high resistance during the OFF-state, and offers least resistance during the ON-state. MN1 transistor acts as a two-terminal diode, and passing the logic in such a way that the proposed domino logic overcomes the issue of charge sharing between the capacitances.

Pre-charge and evaluation are the two operating phases for the domino logic. The operation of these two phases depends on the nature of the used clock signal. The clock (Clk) signal level is low for pre-charge phase, which turns-on MP1 and MP4 transistors. The dynamic node is charged up to the power supply voltage VDD through the MP1, MP3 and MP4 transistors during the pre-charge phase. The input signal for the MP3 transistor is the output of the MN1 transistor. Initially, dynamic node is at logic low, which causes MN1 to turn-OFF and MP3 to turn-ON. It passes the logic from node n1 to node n3. At the start of the pre-charge phase, inputs of the PDN transistors must be at logic low that avoids the discharging of the dynamic node. The proposed domino logic generates the output at logic low in the pre-charge phase.

Evaluation phase of the domino logic starts when Clk switches to logic high. MP1 and MP4 transistors are turned-off, which breaks the conducting path between the power supply to dynamic node. In this case, the inputs of the PDN transistors decide the logic level at dynamic node and produces the output logic accordingly. If PDN transistors are arranged in such a way that a conducting path appears between the dynamic node and ground terminal, then charge at dynamic node is completely dis-charged to ground, and output terminal will generate logic high. If PDN transistors are connected in such a way that a conducting path does not appear between the dynamic node and ground

terminal, then charge stored at dynamic node remains unchanged.

OR logic is implemented using the proposed method. The input-output characteristics for 2-input OR logic is verified for the proposed domino method. The circuit is simulated at 7 nm technology node using Cadence's tools at room temperature. Predictive technology model (PTM) files for multi-gate (MG) devices are utilized at 0.7 V power supply voltage. The assumed clock frequency is 50 Mhz. Fig. 10 illustrates the input-output characteristics for 2-input OR logic circuit.

It can be observed from the Fig. 10 that the proposed method generates the accurate OR output, and can be used for the implementation of the domino logic circuits.

#### 4. Simulation results and discussion

In this paper, authors have evaluated and compared the results of the various leakage reduction domino techniques with the help of Cadence's Virtuoso tool, and MG PTM model files. Authors have added PTM library externally for the analysis of the FinFET circuits in the Cadence tool. This model file contains two type of devices like low standby power (LSTP), and high performance (HP) for any particular application. In this paper, authors have considered SG FinFET in which both gate terminals are shorted together to provide high drive current, and LSTP devices to provide the low power dissipation. Authors have simulated various leakage reduction techniques using simulation program with integrated circuit emphasis (SPICE) code, and then these SPICE codes are simulated using Spectre circuit simulator. Authors have analysed the various performance parameters at room temperature for 0.7 V power supply voltage at 7 nm technology node. The fin of the FinFET device has a height of 18 nm, and a thickness of 5 nm at 7 nm technology node. The presumed number of fins for the FinFET devices is 1. Same parameter assumptions are considered for all the leakage reduction techniques for the unbiased comparison.

##### 4.1. Leakage power dissipation

To evaluate the performance of any portable electronic system, leakage power dissipation is the vital parameter in ultra-DSM regime. Power dissipation of leakage reduction techniques decides the effectiveness of the circuits. Authors have evaluated the power dissipation of the proposed and the existing techniques. Main cause of the power dissipation in electronic circuits is OFF-state leakage current. The impact of the leakage current becomes very severe in ultra-DSM regime, which starts to affect the performance of the circuits. Leakage power dissipation is calculated by the product of the leakage current and power supply as shown in Eq. (1). The comparisons of the power dissipation, propagation delay and PDP for the domino buffer, NOT, 2-input OR, and AND circuits are shown in Table 1.

It can be observed from the Table 1 that the proposed method mitigates large leakage power dissipation followed by the LECTOR technique. The proposed approach reduces power dissipation by 41.63% for the buffer circuit as compared to the best reported existing LECTOR technique. LECTOR technique generates the improper input-output waveform because of the inadequate paths for the charging and discharging of the intermediate terminals. Table 1 also depicts that the AND logic saves the large leakage power dissipation as compared to the OR logic for the different domino methods. In AND logic, PDN consists series connected transistors which forms transistors stack, and helps to remove the large power dissipation. The proposed domino method reduces 52.82% leakage power dissipation for the 2-input AND gate as compared to the best reported existing LECTOR technique.

##### 4.2. Propagation delay

Nowadays, speed of the electronic devices is also the demand of the customers. Propagation delay is the main performance parameter which decides the speed of the circuits. Propagation delay is the best option to

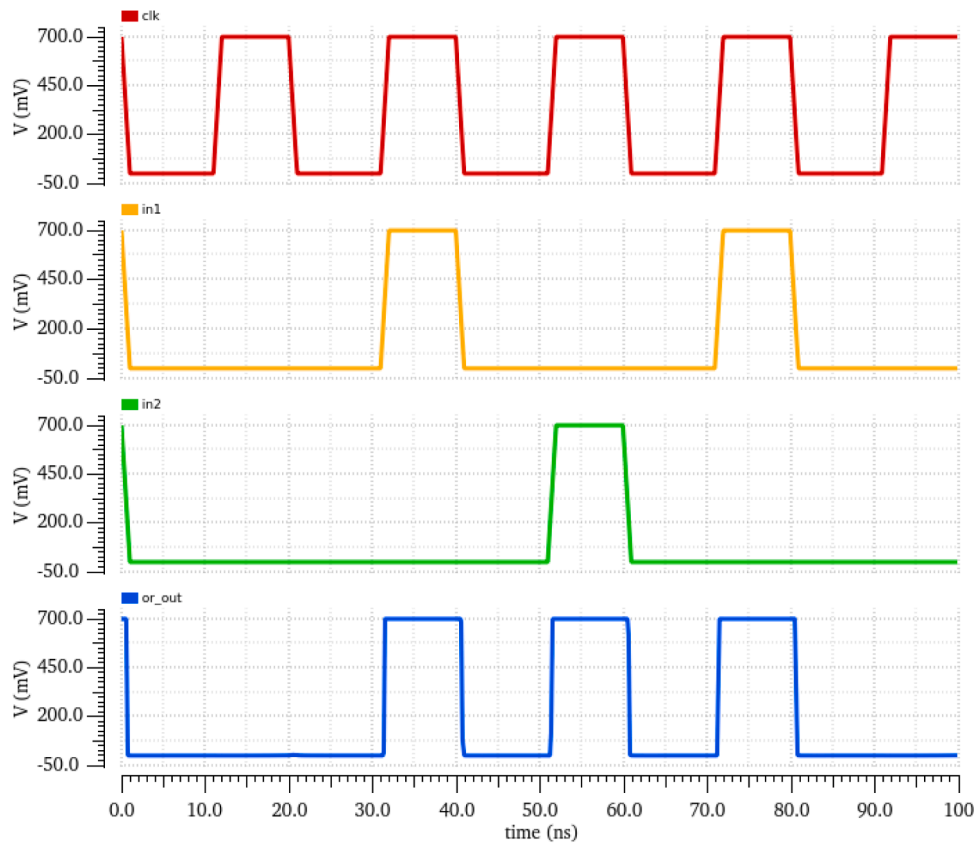


Fig. 10. Input-output characteristics for the proposed low power domino method.

**Table 1**  
Simulation results for power dissipation, propagation delay and PDP for different logic circuits.

| Method                  | Power dissipation (nW) | Propagation delay (ps) | PDP (aJ) |
|-------------------------|------------------------|------------------------|----------|
| LECTOR buffer           | 2.33                   | 178.04                 | 0.41     |
| ONOFIC pull-up buffer   | 3.41                   | 171.13                 | 0.58     |
| Alternate ONOFIC buffer | 3.27                   | 402.47                 | 1.32     |
| DNDFTDL buffer          | 5.91                   | 508.71                 | 3.01     |
| FDSTDL buffer           | 5.60                   | 438.24                 | 2.45     |
| Proposed buffer         | 1.36                   | 79.32                  | 0.11     |
| LECTOR NOT              | 4.81                   | 235.17                 | 1.13     |
| ONOFIC pull-up NOT      | 5.93                   | 172.50                 | 1.02     |
| Alternate ONOFIC NOT    | 5.92                   | 683.62                 | 4.05     |
| DNDFTDL NOT             | 8.37                   | 718.35                 | 6.01     |
| FDSTDL NOT              | 8.11                   | 703.47                 | 5.71     |
| Proposed NOT            | 3.08                   | 86.61                  | 0.27     |
| LECTOR OR               | 2.63                   | 190.14                 | 0.50     |
| ONOFIC pull-up OR       | 3.76                   | 186.31                 | 0.70     |
| Alternate ONOFIC OR     | 3.74                   | 492.30                 | 1.84     |
| DNDFTDL OR              | 6.19                   | 596.83                 | 3.69     |
| FDSTDL OR               | 5.93                   | 511.22                 | 3.03     |
| Proposed OR             | 1.70                   | 92.01                  | 0.16     |
| LECTOR AND              | 1.95                   | 233.06                 | 0.45     |
| ONOFIC pull-up AND      | 2.26                   | 121.34                 | 0.27     |
| Alternate ONOFIC AND    | 2.20                   | 552.64                 | 1.22     |
| DNDFTDL AND             | 3.85                   | 670.28                 | 2.58     |
| FDSTDL AND              | 2.33                   | 593.36                 | 1.38     |
| Proposed AND            | 0.92                   | 138.72                 | 0.13     |

check the performance of the circuits while evaluating the leakage reduction technique because leakage power dissipation is saved at the cost of the decreased speed. If the circuit has a minimal number of transistors then it can respond quickly. Extra transistors are inserted in all the leakage mitigation techniques which induces significant delay. The time difference between the 50% transitions of the output and input signals is used to estimate the delay. Propagation delay is calculated by using the Eq. (9).

$$\tau_p = \frac{\tau_{phl} + \tau_{plh}}{2} \quad (9)$$

Where,  $\tau_p$  is the average propagation delay. Eq. (9) represents average propagation delay in which time difference between 50% levels of the rising input and falling output is  $\tau_{phl}$ , while the time difference between 50% levels of the falling input and rising output is  $\tau_{plh}$ .

Propagation delays for the various logic circuits using various leakage reduction techniques at 7 nm are given in Table 1. The proposed domino approach has less propagation delay than the existing methods for the different logic circuits. The transistor's arrangements in a logic circuit decides the delay value. If arrangement of the transistors are in such a way that logic is flowing easily then it exhibits minimal delay. The proposed domino method decreases the propagation delay by 53.65% as compared to the best reported existing ONOFIC pull-up technique for the buffer circuit. Table 1 is also showing that the propagation delay value for the OR circuit is less as compared to the AND circuit because PDN transistors are connected in parallel in OR circuit. Therefore, logic has small path to flow between the PDN terminals. The speed of the proposed 2-input domino OR circuit is improved by 50.61% as compared to the best reported existing ONOFIC pull-up method.

#### 4.3. Power delay product (PDP)

Domino circuits are used for large memories and high-speed pro-

processors, so the PDP is used to evaluate the energy efficiency. The product of power dissipation and propagation delay is called PDP. PDP must be kept to a minimum value in order to perform well. The approximation of the PDP per switching operation is calculated as shown in Eq. (10).

$$PDP = \text{Power dissipation} \times \text{delay} \quad (10)$$

Table 1 shows the PDP values for the different low power domino logic approaches.

When compared to the existing domino logic approaches, the proposed low power domino method is more capable to reduce the large power dissipation with minimal delay penalty. It improves the PDP metric. The proposed method enhances the PDP metric by 3.78X for the NOT circuit as compared to the best reported existing ONOFIC pull-up approach.

#### 4.4. Temperature variation

Electronic circuits may operate at different temperatures. Therefore, the temperature variation effect on the logic circuits is an important consideration to evaluate a circuit's efficiency [25]. Authors have simulated domino FinFET OR logic using different methods at different temperatures ranging from  $-30^{\circ}\text{C}$  to  $50^{\circ}\text{C}$  to assess the temperature variation effect. The effect of temperature variation on leakage power dissipation is depicted in Fig. 11. Power dissipation is measured for the OR logic for the input combination "11" at 7 nm technology node.

Leakage current depends on the thermal equivalent voltage. Thermal equivalent voltage is directly related to the temperature, and it can be seen from the Eq. (2). Fig. 11 shows that leakage power dissipation increases with the temperature increment. The power dissipation values for the ONOFIC pull-up and alternate ONOFIC domino logics are very close to each other. The power dissipation values for the DNDFTDL and FDSTDL methods are also close to each other. The proposed domino OR logic has less power dissipation fluctuation when temperature rises as compared to other existing domino OR logics. The proposed domino method saves 42.38% power dissipation at  $50^{\circ}\text{C}$  temperature as compared to the best reported existing LECTOR technique.

#### 4.5. Impact of multiple parallel fins

A FinFET device has a fin between the source and drain terminals for the current conduction. To maximize the overall drive current of a FinFET, multiple parallel fins may be fabricated between the source and drain terminals. The number of fins ( $n_{fin}$ ) has a significant effect on the circuit's efficiency, and reliability. FinFET device with multiple parallel fins decreases the inversion charge by generating the charge repulsion between the fins. This increases the hot carrier injection and coupling effect between the fins in the ultra-DSM regime, and creates a problem [26]. The effect of the multiple parallel fins is analyzed for the proposed

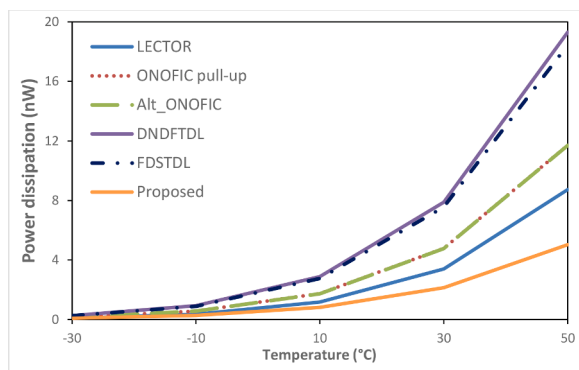


Fig. 11. Temperature variation effect on power dissipation for domino OR logic.

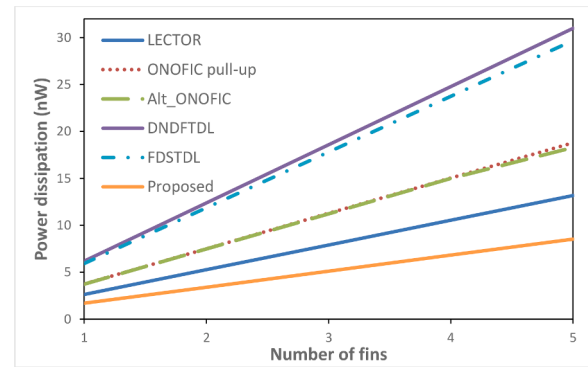


Fig. 12. Parallel fin variation effect on power dissipation for domino OR logic.

method and compared with the other available methods for the domino FinFET OR2 logic. Fig. 12 illustrates the impact of multiple parallel fins varying from 1 to 5 on the leakage power dissipation for the different domino OR logic.

The power dissipation is estimated for the OR logic for the input combination "11". It can be easily observed from the Fig. 12 that the power dissipation for the circuits is increasing with the increment in number of fins. The proposed method shows less variations as compared to the other available methods. It mitigates 72.47% more power dissipation as compared to DNDFTDL method when number of fins is 5.

#### 4.6. Unity noise margin

UNG metric is used to evaluate the noise aware capability of the domino circuits as domino circuits are subjected to the noise. Noise aware circuits must be designed for ultra-low power applications. For the measurement of UNG, a small amplitude pulse with a constant pulse width is added to the inputs of the PDN transistors. When both the input ( $V_{in}$ ) and output ( $V_{out}$ ) amplitudes are identical, then input-output response is observed. UNG is the amplitude level and estimated as shown in Eq. (11) [27].

$$UNG = \{V_{in} : V_{pulse} = V_{out}\} \quad (11)$$

UNG values for the different domino methods for the OR logic are given in Table 2. Large value of UNG reflects the more noise aware circuit.

Table 2 reveals that the proposed method has the highest UNG as compared to other domino OR logics. The proposed low power domino method can be used to design effective domino logic circuits because it has a higher noise tolerance capability than the others.

## 5. Conclusion

Low power dissipation, high speed and scalability are the important features of the FinFET device. The impact of the leakage power dissipation starts to degrade the circuit performance when FinFET technology moves towards the advancement. Leakage reduction techniques are needed to mitigate the impact of leakage power dissipation in VLSI

Table 2

UNG values for different domino methods for OR logic.

| Method           | UNG  |
|------------------|------|
| LECTOR           | 0.47 |
| ONOFIC pull-up   | 0.48 |
| Alternate ONOFIC | 0.47 |
| DNDFTDL          | 0.40 |
| FDSTDL           | 0.41 |
| Proposed         | 0.55 |

technology. Domino logic design pays an attention in large memories and high-speed processors. These circuits are more area efficient as compared to conventional CMOS circuits. In this paper, a circuit level, fixed configuration and single threshold low power domino method is proposed. The proposed method is verified for different domino logic circuits at 7 nm technology node for the leakage power dissipation, propagation delay and PDP. The impact of the temperature variation, multiple parallel fins variation and UNG are evaluated to check the reliability of the circuits. The effect of the temperature, number of fins variations and UNG values are observed for the OR logic for the different domino methods. MG PTM LSTP FinFET model files are utilized for the circuit simulations. **The proposed low power domino method reduces the power dissipation by 72.54% and minimizes the propagation delay by 84.58% for the OR logic as compared to the DNDFTDL domino technique. The proposed domino method is 95.66% more energy efficient and 27.27% more noise tolerance for the OR logic as compared to the DNDFTDL domino technique.**

### CRedit authorship contribution statement

: Data curation, Formal analysis, Investigation, Validation, Writing – original draft. **Vijay Kumar Sharma:** Conceptualization, Methodology, Resources, Supervision, Writing – review & editing.

### Declaration of Competing Interest

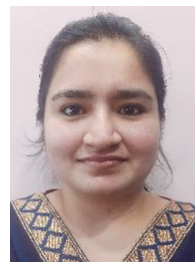
The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

### Data availability

No data was used for the research described in the article.

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